

Dynamic Programming Trajectory Optimization and its application to D-SEND #2 Low Sonic-boom Research Project

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JAXA develops an unmanned technology demonstrator vehicle for low sonic boom research. The research project is called D-SEND #2 (Drop test for Simplified Evaluation of Non-symmetrically Distributed sonic boom). The autonomous demonstrator vehicle will be dropped from a high altitude balloon, and it will accelerate and glide to fly over a predetermined measurement area with a supersonic cruise condition. Trajectory optimization analysis is necessary for the design of onboard reference trajectory generation logic, which is robust against the dropping point uncertainty. An easy-to-use trajectory optimization tool using dynamic programming approach has been developed in order to generate reference trajectories in the early stage of design process. This paper introduces how the dynamic programming approach is applied to the D-SEND #2 flight trajectory optimization, and it shows numerical examples, which include the maximum and minimum range problems, and range adjustment problems.

I. Introduction

Aviation Program Group, Japan Aerospace Exploration Agency (JAXA) performs the D-SEND (Drop test for Simplified Evaluation of Non-symmetrically Distributed sonic boom) project for low sonic boom research. The project is composed of D-SEND #1 and D-SEND #2. D-SEND #1 successfully completed the drop test in 2011, where two different axisymmetric bodies were dropped from a high altitude balloon and the sonic booms were measured and compared with each other. The experimental results clearly showed reduction in sonic boom by low sonic boom design technology¹. JAXA is currently preparing the D-SEND #2, where an experimental supersonic airplane model based on JAXA's proprietary low sonic boom design technology is dropped from a high altitude balloon and the sonic boom is measured, as shown in the illustration of Fig. 1. The sonic booms are measured and recorded by the microphone system that is tied to a line between the ground and the blimp at an altitude of 1 km. The drop test will take place at the Esrange Test Site near Kiruna, Sweden in 2013.

The vehicle is fully autonomous to perform supersonic flight with a prescribed lift at a measurement area located above the ground measurement station. The guidance and control system design, which includes the onboard logic for reference trajectory generation, is one of the critical items in the vehicle design. It should consider mission requirements for the sonic boom measurement, not only for the final supersonic flight condition, but also mid-term supersonic flight. Range adjustment against uncertainty of the dropping point is a challenge for the design. Since the amount of range adjustment could enlarge permissible dropping area, it increases the probability of mission success. Since the vehicle has no recovery device for reuse, the success probability has to be maximized².

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In order to support the reference trajectory generation logic design, results of the trajectory optimization should be useful, but in general, solving the optimal control problem requires large amount of engineering resources as well as computational resources. One of the reasons is that there is no easy-to-use tool which can be applied in various kinds of optimization problems. Selection from optimization tools, preparing data including modeling and performance index, and setting adjustable parameters for each tool are not a fool-proof process, but they require some level of experience, or they tend to be a time consuming trial and error process. Therefore, in the real design, the reference trajectory generation logic design heavily depends on engineer's know-how and experience. High Speed Flight Demonstration Phase 2 (HSFD-II) is a typical case, where the reference trajectory from the balloon drop to deployment of the parachute is designed based on the know-how of glider pilots and simple geometrical calculations³⁻⁵. Since authors were involved in the HSFD-II GNC development, their sheer experience is a strong motivation for this research to incorporate an optimization tool in the design process.

Kyushu University participates in the project to support developing a computational tool for trajectory optimization by the method of Dynamic Programming (DP). This paper is a product from the collaboration between Kyushu University and JAXA D-SEND #2 project Guidance, Navigation and Control (GNC) design team. Although the research and development are aimed at the project, the result is useful for other applications of DP.

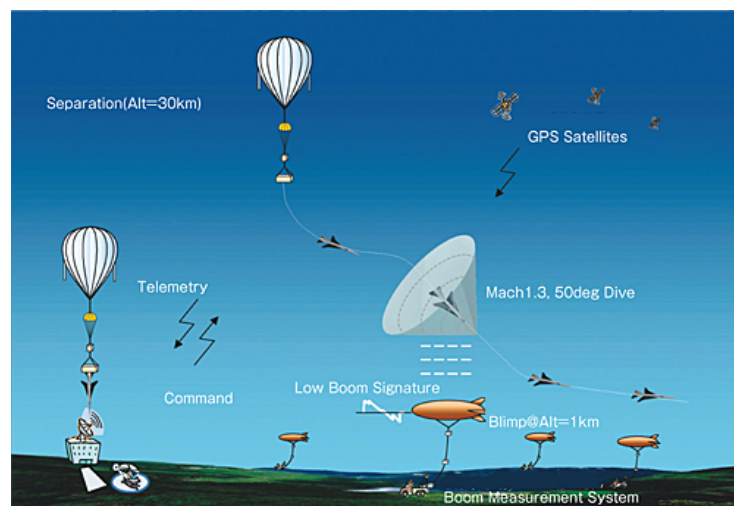


Figure 1. Engineering image of the D-SEND #2 flight experiment

II. Dynamic Programming

Before the discussion of concrete application, characteristics of DP are summarized in this chapter. Needless to say, there are many numerical approaches for vehicle trajectory optimization problems. Among them, methods of solving the Euler-Lagrange differential equation and those based on parametric presentation and nonlinear programming are commonly used. Although they could provide accurate solutions, they require time consuming process due to their natures, such as, iterative calculations to be converged are necessary, solutions satisfying the necessary conditions are a local minimum, and programming is inevitably complex in order to include equality and inequality constraints. On the other hand, dynamic programming, which was proposed by Richard Bellman in 1950s based on the Hamilton-Jacobi-Bellman equation has the following advantages⁶⁻⁹.

- 1) Global optimization: DP defines a grid in the search space of state variables, and it finds the optimal solution from every possible combination of transitions. Therefore, it gives the global optimum, not a local solution that satisfies the necessary condition.
- 2) No iterative computation for convergence and computational time predictable in advance: Computational process in DP is decisive and it does not contain any iterative computation for the optimal solution. Therefore, it can estimate the computational time in advance, which depends on the number of grid points in the state space.
- 3) No difficulty in including inequality constraints: DP defines a grid in the state space. Therefore, it is easy to include inequality constraints on the state variables themselves and their functions. Rather, inequality constraints are suitable to make the search space limited. Inequality constraints on the control variables are also easy to be implemented.

4) Simplicity of computer program: Discrete form of the Hamilton-Jacobi-Bellman optimality condition makes the computer program simple and easy to understand.

On the other hand, DP has the following disadvantages, which generally affects its usage in engineering design.

1) Curse of dimensionality: Computational time and memory expand rapidly against the number of state variables. Currently three dimensions would be a limit as a design tool when a moderate personal computer is used.

2) Quantization error: DP finds the optimal solution from every combination of transitions of grid points in the state space. Accuracy of the solution depends on the grid, and it often needs some adjustment to make the solution accurate enough to be useful for practical purpose. Furthermore, if the number of control variables is not enough to realize the transition between the two grid points, it requires fine grid and large computational time.

Since the dimension of the trajectory optimization for D-SEND #2 is three, the computational time due to curse of dimensionality doesn't make a considerable influence. Concerning the quantization problem, D-SEND #2 longitudinal motion is the case that the number of control variables is only one against three state variables. This paper has proposed a technique to avoid fine grid by augmenting an artificial control variable.

III. D-SEND #2 Project and mission requirement

Fig. 2 shows an illustration of the vehicle. The unpowered model vehicle has the following dimensions. The body length is 7.7 m, the span is 3.5 m and the wing area is 4.89 m². The mass is one ton (1000 kg).

The top priority of the mission requirement is to simulate the supersonic cruise flight and atmospheric propagation of the sonic boom in order to evaluate the sonic boom reduction research result. It means that the terminal flight condition above the ground measurement station should satisfy the prescribed conditions imposed on mach number, lift coefficient, flight path angle and altitude. On the other hand, the mid-term flight from the balloon separation to the terminal point, is basically free but it should satisfy inequality constraints on variables, such as, mach number, load factor, equivalent air speed, bank angle, angle of attack, control surfaces angles, and so on. By using the freedom of flight velocity and glide slope angle in the mid-term flight, it can adjust the flight range of the vehicle. The range adjustment introduces permissible area for the separation point which depends on the altitude. If the permissible area increases, the mission success probability would be higher.

In the logic design of the reference trajectory, solutions of the optimal control problem are expected to be helpful. For example, the maximum range is the most important concern, because it gives the upper bound for permissible separation point from the ground measurement station. The minimum range depending on the flight profile is another concern. As discussed in Introduction, however, numerical computations for the optimal solution with the actual vehicle's dynamical system are generally not simple, but it often needs exhaustive time and engineering resources. The DP method is chosen to overcome the situation. The following chapters will show how it is applied to trajectory optimization in the D-SEND #2 project.



Figure 2. D-SEND #2 low sonic boom vehicle.

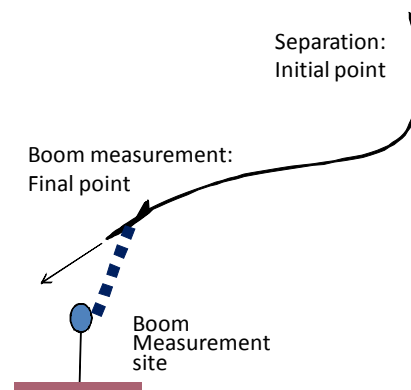


Figure 3. Sketch for the D-SEND #2 flight trajectory.

IV. Dynamics of D-SEND #2 vehicle

It assumes that the rotational motion of the vehicle is so fast that the flight trajectory can be described by translational motion of a point mass. Dynamics of the vehicle can be presented by the following differential equations.

$$\frac{dV}{dt} = -g \sin \gamma - D/m \quad (1)$$

$$V \left(\frac{d\gamma}{dt} - \frac{V \cos \gamma}{R_o + H} \right) = -g \cos \gamma + L \cos \sigma / m \quad (2)$$

$$\frac{dH}{dt} = V \sin \gamma \quad (3)$$

$$\frac{d\phi}{dt} = \frac{V \cos \gamma}{R_o + H} \quad (4)$$

$$\frac{d\psi}{dt} = \frac{L \sin \sigma}{mV \cos \gamma} \quad (5)$$

Velocity vector is presented by true air speed V , flight path angle γ , and heading angle ψ . Position is presented by altitude H and earth center angle ϕ for flight range. L and D are lift and drag, respectively. σ is bank angle, i.e. angle of rotation for lift around velocity vector. m , g and R_o are mass of the vehicle, gravitational acceleration and earth radius, respectively. Lift and bank angle are control variables. The aerodynamics for translational motion can be calculated for zero-moment trimmed conditions by giving a center-of-gravity position. The lift L and drag D are functions of air density ρ , true air speed V , angle of attack α , Mach number M , and altitude H as follows.

$$L = \frac{1}{2} \rho(H) V^2 S C_L(\alpha, M, H) \quad (6)$$

$$D = \frac{1}{2} \rho(H) V^2 S C_D(\alpha, M, H) \quad (7)$$

where S is wing area of the vehicle. The lift and drag coefficients C_L, C_D are given by the aerodynamic data table, which are generated by the CFD analysis. The trimmed aerodynamic data are presented by the following approximation in order to simplify the computation, where the coefficients $C_{D_o}(M, H)$ and $K(M, H)$ are functions of Mach number and altitude.

$$C_D = C_{D_o}(M, H) + K(M, H) C_L^2 \quad (8)$$

The angle of attack has upper and lower bounds, which give upper and lower limitations on the lift coefficient.

$$C_{L_{min}}(M, H) \leq C_L \leq C_{L_{max}}(M, H) \quad (9)$$

The aerodynamics table is given for the following discrete grid points, where linear interpolation is performed to reconstruct the aerodynamic coefficient for an arbitrary series of Mach number and altitude.

$$M = [0.2, 0.4, 0.6, 0.8, 0.9, 0.95, 1.05, 1.1, 1.2, 1.4, 1.6]$$

$$H = [0, 2000, 4000, \dots, 32000, 34000] \text{ [m]}$$

Among the five state variables in Eqs. (1) to (5), flight range, i.e. earth center angle ϕ , and heading angle of the velocity vector ψ do not give any influence on other state variables, so they can be used in the performance index by time integral form, which is to be defined in each optimization problem. Three remained variables, that is, V , γ , H are the state variables of concern.

Initial and final conditions are given for the state variables as follows. Flight path angle at the final point can be free in order to analyze the effect of prescribing the condition.

$$\gamma(t_o) = \gamma_o, \quad V(t_o) = V_o, \quad H(t_o) = H_o \quad (10)$$

$$\gamma(t_f) = \gamma_f, \quad V(t_f) = V_f, \quad H(t_f) = H_f \quad (11)$$

Reduction of the system order by replacing independent variable

Since the state equation does not contain any time dependent parameters, the independent variable “time” can be replaced by a state variable. The replacement can reduce the order of the state equation by one, where order

reduction is beneficial for saving the DP computational time. In order to apply the optimality condition by the DP method, the new independent variable should be a monotonous function of the time. Altitude is a candidate for the independent variable, but it is necessary to put an inequality constraint on glide path angle. In case of unpowered vehicles like the D-SEND#2 demonstrator, however, the following total energy divided by mass, E^* is suitable, because it continuously decreases with time.

$$E^* = \frac{1}{2}V^2 + g_o H_p \quad (12)$$

where g_o ($=9.80665 \text{ m/s}^2$) is the standard gravitational acceleration, and H_p is geopotential altitude defined by the following.

$$H_p = \frac{H}{1 + H/R_o} \quad (13)$$

Time derivative of the new independent variable gives reduced order state equations.

$$\frac{dE^*}{dt} = V \frac{dV}{dt} + g_o \frac{dH_p}{dt} = \frac{-DV}{m} \quad (14)$$

It introduces dV/dE^* and $d\gamma/dE^*$ as follows.

$$\frac{dV}{dE^*} = \frac{dV}{dt} / \frac{dE^*}{dt} = \frac{1}{V} \left(1 + \frac{mg \sin \gamma}{D} \right) \quad (15)$$

$$\frac{d\gamma}{dE^*} = \frac{d\gamma}{dt} / \frac{dE^*}{dt} = \frac{mg \cos \gamma - L \cos \sigma - \left(\frac{mV^2 \cos \gamma}{R_o + H} \right)}{V^2 D} \quad (16)$$

Other variables necessary for performance index calculation is as follows.

$$\frac{d\phi}{dE^*} = \frac{d\phi}{dt} / \frac{dE^*}{dt} = -\frac{\cos \gamma}{R_o + H} \left(\frac{m}{D} \right) \quad (17)$$

$$\frac{d\psi}{dE^*} = \frac{d\psi}{dt} / \frac{dE^*}{dt} = -\frac{L}{D} \frac{\sin \sigma}{V^2 \cos \gamma} \quad (18)$$

Since the altitude is explicitly expressed by the independent variable E^* and velocity V , Eq. (3) can be excluded from the state equation.

V. Maximum range problem

In the beginning, the maximum range problem is discussed. The performance index is flight range which is written by the following integral equation.

$$J = R_o \phi(t_f) = \int_{t_o}^{t_f} \frac{V \cos \gamma}{1 + H/R_o} dt = - \int_{E_o^*}^{E_f^*} \frac{m \cos \gamma}{D(1 + H/R_o)} dE^* \quad (19)$$

The independent variable E^* is the total energy divided by the vehicle mass. It is obvious for the maximum range problem that the lateral motion should be suppressed in case of no wind conditions, i.e. bank angle is null. The state variables are velocity V and flight path angle γ .

It is necessary for the DP optimization to define a grid in the space of the independent variable and the state variables. Here, the most simple orthogonal iso-grid is used for the three dimensional space of (E^*, V, γ) . DP trajectory optimization is carried out by assuming that the trajectory consists of a series of local transitions between two points on each neighboring energy E_k^* and E_{k+1}^* . If the numbers of each division for (E^*, V, γ) are defined as N_E, N_V, N_γ , the total number of combinations for the possible paths becomes $(N_V \times N_\gamma)^{N_E}$, which is so huge number that it is impossible to calculate every combination. DP can solve the problem by using optimality condition for each local transition between two points on each neighboring energy, which can be written as follows.

$$J_{opt}(E_k^*, V_{k_i}, \gamma_{k_j}) = \max_{V_{k+1_i} \rightarrow V_{k_i}, \gamma_{k+1_j} \rightarrow \gamma_{k_j}} \left\{ \Delta J(\Delta E^*, V, \gamma) + J_{opt}(E_{k+1}^*, V_{k+1_i}, \gamma_{k+1_j}) \right\} \quad (20)$$

where the fraction of the performance index at each transition is given by,

$$\Delta J = - \int_{E_{k+1}^*}^{E_k^*} \frac{m \cos \gamma}{D(1 + H/R_o)} dE^* \quad (21)$$

As shown in Fig. (4), every transition of grid points on (V, γ) between the neighboring energy space $E_{k+1}^* \rightarrow E_k^*$, right hand side of Eq. (20) is maximized to introduce $J_{opt}(E_k^*, V_{k_i}, \gamma_{k_j})$.

At the initial point, the performance index is given.

$$J_{opt} = J_{opt_o} = 0 \quad (22)$$

$J_{opt}(E_k^*, V_{k_i}, \gamma_{k_j})$ can be calculated by the optimality condition Eq. (20), from the initial energy to the final, $E_o^* \rightarrow E_f^*$, i.e. from right to left in Fig. 4. Thanks to the optimality condition, total number of cases to calculate the transition can be reduced to $(N_V \times N_\gamma)^2 \times N_E$.

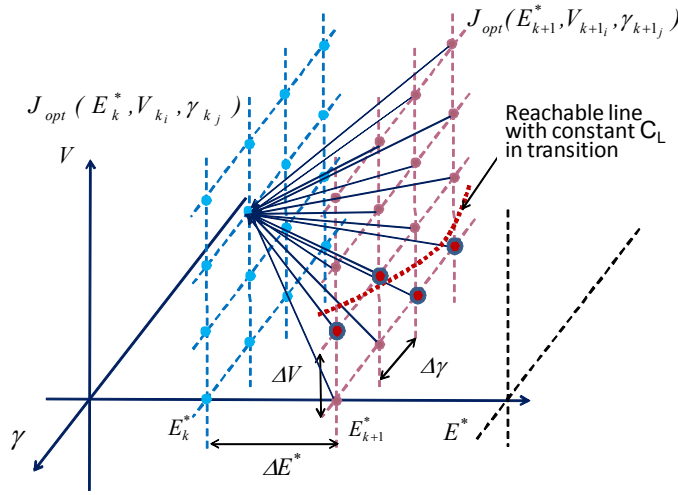


Figure 4. Grid and calculation of the optimal trajectory by DP.

Controllability for the transition between two grid points

Lift, or Lift coefficient, is only a control variable for longitudinal motion of the vehicle. Transition between two grid points gives two equations for the change of velocity and flight path angle. As the transition obtained by changing lift coefficient assumed constant in the transition, which is shown schematically in Fig. 4, it is difficult to connect two grid points by a single control variable. Even if it is relaxed to permit some equation errors, it should require a fine grid, which introduces a large computational time. Here, a technique that uses an artificially augmented control variable is proposed to solve this difficulty. The artificially augmented control variable considered is additional drag coefficient put on the original drag coefficient that is a function of the lift coefficient as Eq. (8). Since the artificial variable is additional drag, it gives negative effect on maximizing range, therefore the amount of additional drag can be minimized, and it gives a conservative solution of the maximum range. This technique can be called “augmented control variable” (ACV) method. The detailed calculation is as follows.

Eqs. (15) and (16) lead the following equations of discrete form.

$$\frac{\Delta V}{\Delta E^*} = \frac{1}{V} \left(1 + \frac{mg \sin \gamma}{D} \right) \quad (23)$$

where $\Delta E^* = E_2^* - E_1^*$, $\Delta V = V_2 - V_1$, $V = (V_1 + V_2)/2$, $\gamma = (\gamma_1 + \gamma_2)/2$

Eq. (23) introduces drag coefficient, that is $\Delta V / \Delta E^* \rightarrow D \rightarrow C_D$

$$\frac{\Delta\gamma}{\Delta E^*} = \frac{mg \cos \gamma - L - (\frac{mV^2 \cos \gamma}{R_o + H})}{V^2 D} \quad (24)$$

where $\Delta\gamma = \gamma_2 - \gamma_1$.

Eq. (24) generates lift coefficient, that is $\Delta\gamma / \Delta E^* \rightarrow L \rightarrow C_L$, and it should satisfy the inequality constraint of Eq. (9).

Actual drag coefficient C_D depends on the lift coefficient C_L as defined by Eq. (8). C_D^* derived by Eq. (23) is augmented control variable presenting additional drag like a speed-brake, which satisfies the following equation. $\Delta C_{D_{max}}$ is a properly given constant.

$$\Delta C_{D_{max}} \geq C_D^* - C_D \geq 0 \quad (25)$$

Performance index fraction between the two grid points is given by the following, where ΔJ is positive in case of decreasing energy, i.e. $\Delta E^* < 0$.

$$\frac{\Delta J}{\Delta E^*} = -\frac{m \cos \gamma}{D(1 + H/R_o)} \quad (26)$$

An example from the numerical results

Numerical results obtained by the developed DP trajectory optimization tool are introduced in this section. It is assumed that initial acceleration to some equivalent air speed is conducted from balloon drop at an altitude of 30km with 20 seconds free fall to align the heading by roll maneuver, and then after a maximum lift coefficient 0.6 is maintained for 28.5 seconds in order to pull up quickly to a path angle of -1 radian. It gives the following initial condition.

$$\begin{aligned} \gamma(t_o) &= -1.0 \text{ [rad]}, \quad V(t_o) = 409 \text{ [m/s]} \\ H(t_o) &= 19,388 \text{ [m]}, \quad J(t_o) = 1,971 \text{ [m]} \end{aligned}$$

The final condition is given as follows to satisfy the sonic boom measurement requirement.

$$V(t_f) = 400 \text{ [m/s]}, \quad H(t_f) = 10,000 \text{ [m]}, \quad (E_f^* = 177,913 \text{ [m}^2/\text{s}^2])$$

The vehicle has the following limitations imposed on Mach number, load factor, lift coefficient, and equivalent air speed.

$$\begin{aligned} 1.1 &\leq M \leq 1.7 \\ \frac{L}{mg} &\leq 3.5 \\ C_{L_{min}}(M, H) &\leq C_L \leq C_{L_{max}}(M, H) \\ V_{EAS} &\leq V_{EAS_{max}} \end{aligned}$$

The equivalent airspeed constraint is generally not active because it becomes the maximum at the final point, or at the lowest altitude. Other three limitations are critical in the trajectory optimization.

Grid points are given as follows. The velocity boundary corresponds to the Mach number limitation.

$$\begin{aligned} \gamma_{min} &\leq \gamma \leq \gamma_{max}, \quad \gamma_{min} = -1.0, \quad \gamma_{max} = 0, \quad \Delta\gamma = 0.005 \text{ (number of division=200)} \\ V_{min} &\leq V \leq V_{max}, \quad V_{min} = 324 \text{ [m/s]}, \quad V_{max} = 502 \text{ [m/s]}, \quad \Delta V = 1 \text{ [m/s]} \text{ (number of division=178)} \\ E_{min}^* &\leq E^* \leq E_{max}^*, \quad E_{min}^* = 177,913 \text{ [m}^2/\text{s}^2], \quad E_{max}^* = 273,237 \text{ [m}^2/\text{s}^2] \\ \Delta E^* &= -3075 \text{ [m}^2/\text{s}^2] \text{ (number of division=31)} \end{aligned}$$

To accelerate the computation by reducing search space, the following conditions are added

$$\begin{aligned} -1.3 \times 10^{-2} &\leq dV / dE^* \leq 1.3 \times 10^{-2} \text{ [s/m]} \\ -5 \times 10^{-5} &\leq d\gamma / dE^* \leq 5 \times 10^{-5} \text{ [s}^2/\text{m}^2] \end{aligned}$$

Figures 5a), 5b), and 5c) show the optimal trajectory for the prescribed terminal path angle case, i.e. -45 degrees. Dot symbols in the figures indicate points given as an initial condition or final condition. Figure 6 shows the maximum range with respect to terminal path angle. The maximum range is obtained at -48 degrees of the final flight path angle. The effect of imposing the flight path angle condition at the final is not so large around the optimum angle.

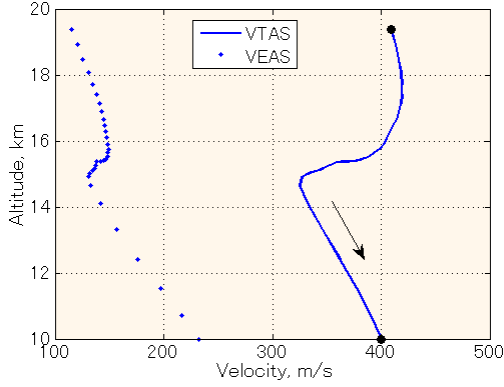


Figure 5a) Maximum range optimal trajectory, velocity final flight path angle=-45 degree.

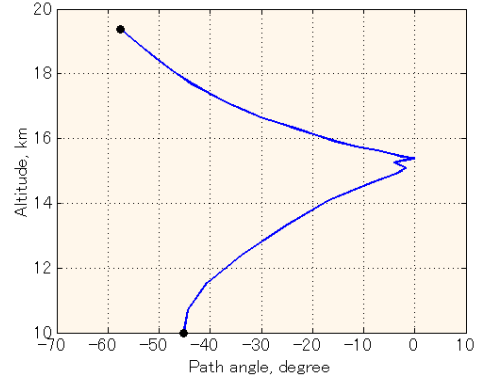


Figure 5b) Maximum range optimal trajectory, flight path angle, final flight path angle=-45 degree.

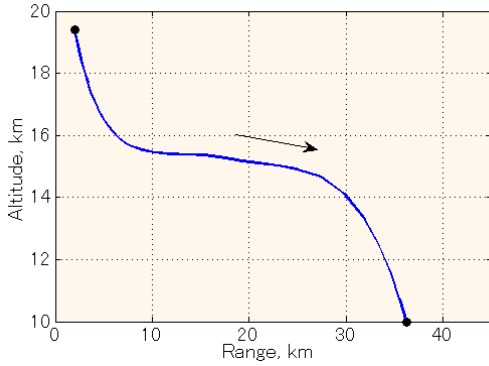


Figure 5c) Maximum range optimal trajectory, range, final flight path angle=-45 degree.

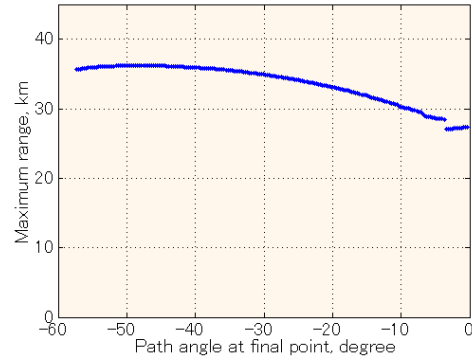


Figure 6. Maximum range with respect to terminal glide path angle.

VI. Minimum range problem

The minimum range is another extreme to be considered. The same performance index as defined by Eq. (19) should be minimized. The same tool for the maximum range problem can be used by changing the property of the augmented artificial control variable added on the drag force. In the previous maximum range problem, positive deviation drag force artificially considered to connect grid points for each transition, which are naturally suppressed in maximizing the range. In the minimum range problem, however, the direction of the drag force should be opposite, i.e. the adjustment force put on the drag should be negative, or thrust. The negative adjustment force to connect the grid points are naturally suppressed in minimizing the range. The following condition is imposed instead of Eq. (25), where ε ($1 > \varepsilon > 0$) is a properly given constant parameter.

$$C_D \geq C_D^* \geq \varepsilon C_D \quad (27)$$

The following is a typical example from the minimum range analysis. The initial condition is defined from the result of free fall of 40 seconds to obtain the minimum range problem. Limitations to be considered are the same as those of previous maximum range example. Grid points are the same, except for the flight path angle and the total energy initial condition.

$$\gamma(t_o) = -\pi/2 \text{ [rad]}, \quad V(t_o) = 384.9 \text{ [m/s]}$$

$$H(t_o) = 23,254 \text{ [m]}, \quad J(t_o) = 0 \text{ [m]}$$

$$\gamma_{\min} \leq \gamma \leq \gamma_{\max}, \quad \gamma_{\min} = -\pi/2, \quad \gamma_{\max} = 0, \quad \Delta\gamma = 0.00785 \text{ (number of division=200)}$$

$$V_{\min} \leq V \leq V_{\max}, \quad V_{\min} = 324 \text{ [m/s]}, \quad V_{\max} = 502 \text{ [m/s]}, \quad \Delta V = 1 \text{ [m/s]} \text{ (number of division=178)}$$

$$E_{\min}^* \leq E^* \leq E_{\max}^*, \quad E_{\min}^* = 177,913 \text{ [m}^2\text{/s}^2\text{]}, \quad E_{\max}^* = 301,289 \text{ [m}^2\text{/s}^2\text{]}$$

$$\Delta E^* = -3,009 \text{ [m}^2\text{/s}^2\text{]} \text{ (number of division=41)}$$

$$\varepsilon = 0.5$$

Figures 7a), 7b), 7c) show the optimal trajectory for the case of free final path angle. The obtained minimum range is about 8 km. The flight continues free fall as long as possible to reach the maximum Mach number 1.7 condition. It pulls up with maintaining the maximum Mach number as long as possible to satisfy the final true air speed 400 m/s. The equivalent air speed reaches about 250 m/s at the maximum.

Figure 8 shows the effect of prescribing the final flight path angle on the minimum range. The minimum range increases with flight path angle. The maximum range obtained in the section V. is also plotted in the figure. Although the adjustable range is maximum around -30 degrees, final flight path angle of -45 degrees is selected by considering the measurement mission requirement. Even in the case, the adjustable range is more than 20 km.

Figures 9a), 9b), 9c) show the optimal trajectory for the case of final flight path of -45 degrees. The flight profile shows that shallow glide is necessary to realize the final flight path angle, which increases the minimum flight range.

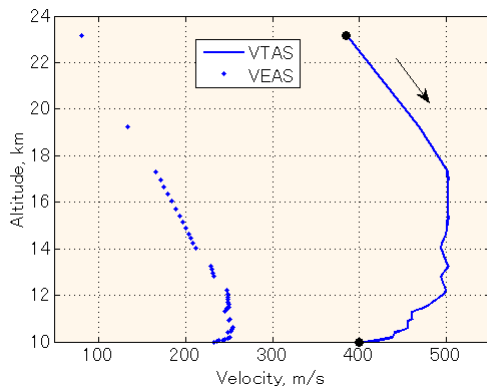


Figure 7a) Minimum range, velocity, free final flight path angle.

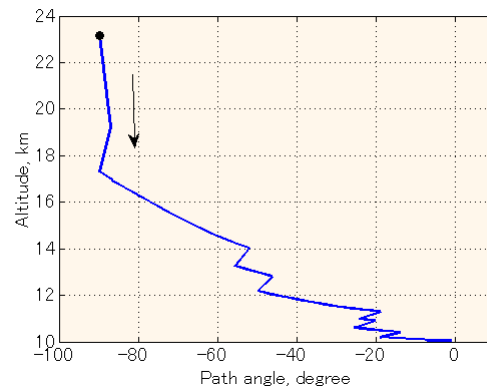


Figure 7b) Minimum range, path angle, free final flight path angle.

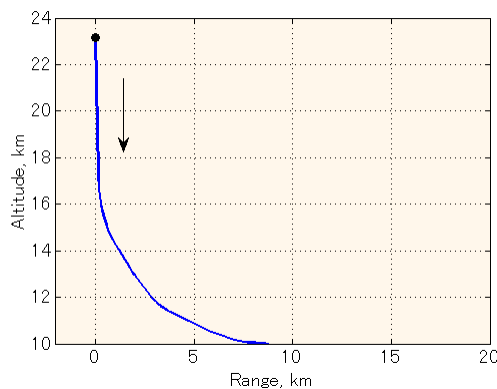


Figure 7c) Minimum range, flight path, free final flight path angle.

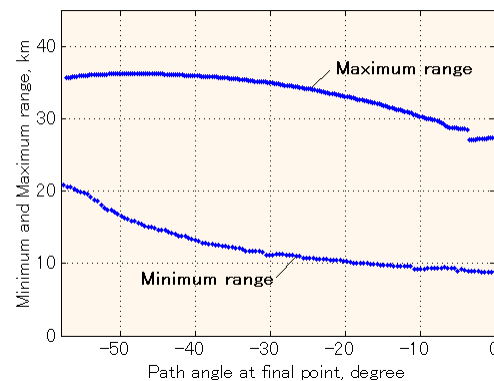


Figure 8. Minimum range with respect to flight path angle at final point.

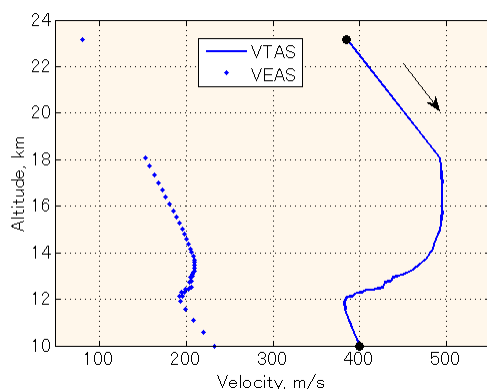


Figure 9a) Minimum range, velocity, final flight path angle=-45 degree.

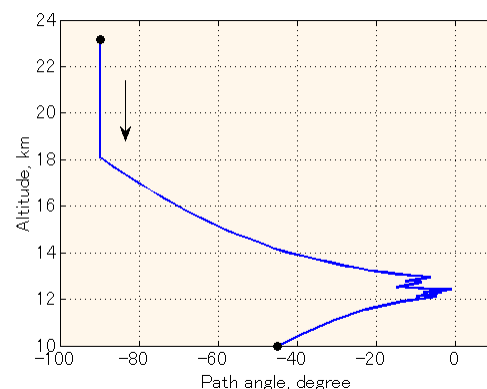


Figure 9b) Minimum range, flight path angle, final flight path angle=-45 degree.

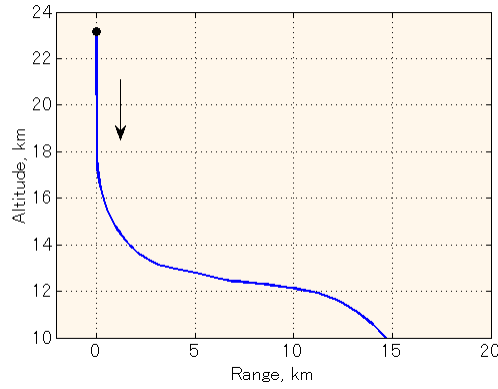


Figure 9c) Minimum range, flight path, final flight path angle=-45 degree.

VII. Range control with longitudinal motion

The tool has demonstrated to generate trajectories of the maximum range and the minimum range of the experiment vehicle. It is assumed that the heading of the initial pull-up after the free fall from the balloon drop is aligned to the measurement site, lateral motion is suppressed by setting the bank angle null. Between the two extreme cases discussed in previous sections, range can be adjusted by trading off an appropriate performance indices. A performance index presenting the following time average of load factor deviation from equilibrium condition can introduce minimization of the vehicle maneuver.

$$J_1 = \int_{t_o}^{t_f} V \left| \frac{d\gamma}{dt} \right| dt = - \int_{E_o^*}^{E_f^*} V \left| \frac{d\gamma}{dE^*} \right| dE^* \quad (28)$$

Trade-off against the range is performed by the following performance index, where a is a weight parameter.

$$J = J_1 + aJ_2 = - \int_{E_o^*}^{E_f^*} V \left| \frac{d\gamma}{dE^*} \right| dE^* - a \int_{E_o^*}^{E_f^*} \frac{m \cos \gamma}{D(1 + H/R_o)} dE^* \quad (29)$$

where the range is as previously defined.

$$J_2 = R_o \phi(t_f) = \int_{t_o}^{t_f} \frac{V \cos \gamma}{1 + H/R_o} dt = - \int_{E_o^*}^{E_f^*} \frac{m \cos \gamma}{D(1 + H/R_o)} dE^* \quad (30)$$

The augmented control variable method proposed in previous sections is not applicable to this performance index. In order to overcome the controllability problem, only one grid point along the velocity change for each flight path change is selected to minimize drag force error in each transition. The method is rational because the flight path angle is directly affected by control variable L , but on the other hand, the velocity is mainly affected by flight path angle γ , one of the two state variables. The approach could be called Least-error Grid-point Selection method (LGS).

Numerical results are shown in the following, where conditions including initial and final points, grid, inequality constraints are the same as those of the maximum range example.

Figures 10a), 10b), 10c) show the optimal trajectory of the zero weight and free terminal path angle case, where the average of load factor deviation is only considered. It pulls up in order to minimize the increase of true airspeed to the equilibrium glide condition of target speed of 400 m/s. After it reaches the glide condition, it continues straight flight to the terminal point. The flight profile is quite smooth to realize the final point condition.

Figures 11a), 11b), 11c) show the optimal trajectory of the zero weight and prescribed terminal path angle of -45 degree case. Two other cases, the maximum range and positive weight of $a=0.3$, are also plotted for comparison. The terminal path angle condition introduces more complicated flight, the flight profile of which is similar to that of the maximum range problem. This case, however, the shallow glide is realized at lower altitude, therefore the flight range is shorter than the maximum range.

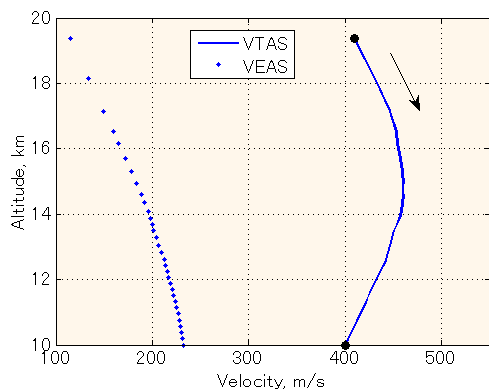


Figure 10a) Minimum average load factor, velocity, free final flight path angle.

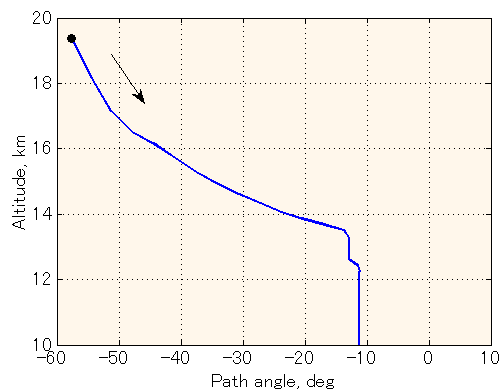


Figure 10b) Minimum average load factor, path angle, free final flight path angle.

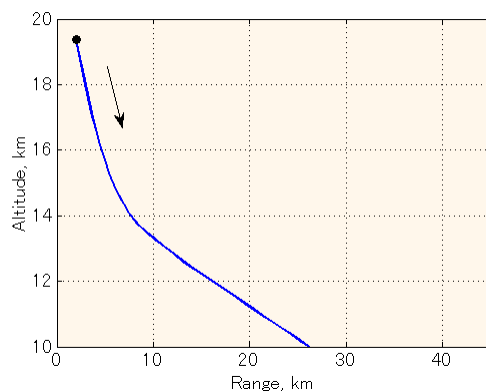


Figure 10c) Minimum average load factor, flight path, free final flight path angle.

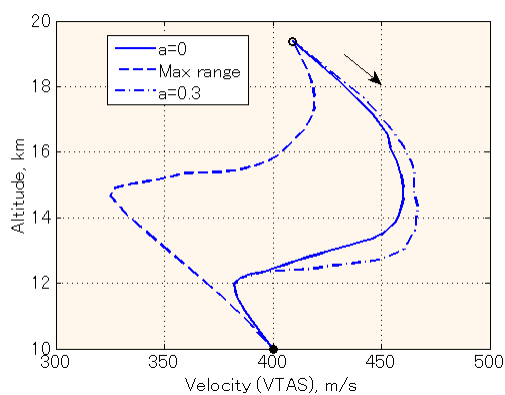


Figure 11a) Range control, velocity, final flight path angle=-45 degrees.

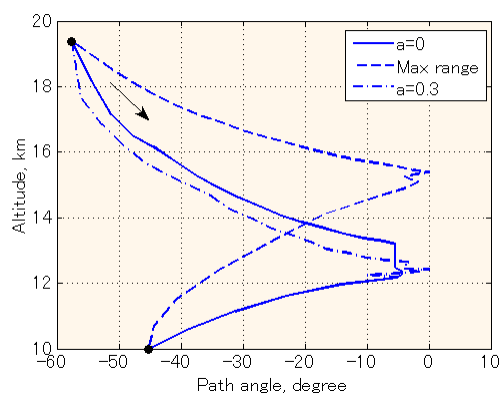


Figure 11b) Range control, path angle, final flight path angle=-45 degrees.

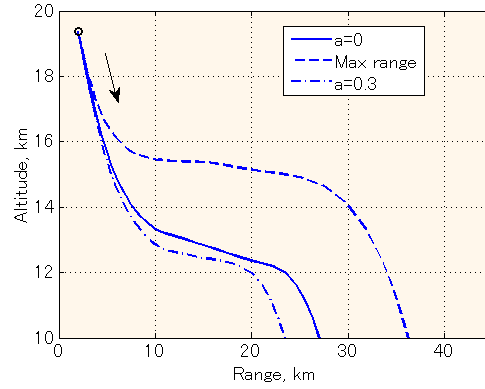


Figure 11c) Range control, flight path, final flight path angle=-45 degrees.

VIII. Range control with lateral motion

Range control by using lateral motion depicted in Fig. 8 is a possible plan. The actual trajectory design excluded the lateral maneuver because it complicates the sonic-boom propagation to the measurement site. The tool, however, has demonstrated its capability to the application. Lateral motion freedom introduces another real control variable, that is bank angle, to the system, therefore, it makes the grid transition easier than the previous single control variable cases. This section shows an example calculated as a reference data for the project.

The following performance index is defined to be maximized, where it is considered that the flight length and turn angle can be a trade-off.

$$J = \int_{E_o^*}^{E_f^*} R_o \frac{d\phi}{dE^*} dE^* + a \int_{E_1^*}^{E_2^*} \frac{d\psi}{dE^*} dE^* = \int_{E_o^*}^{E_f^*} R_o \frac{d\phi}{dE^*} dE^* + a(\psi(E_2^*) - \psi(E_1^*)) \quad (31)$$

The first term is flight length and the second term is total turn angle, where a is a weight variable to balance the two performance indices. Figures 13a) to 13e) are obtained numerical results. The range is reduced to 30km, and straight length from the separation point to the final point becomes about 21 km. The lower boundary of Mach number $M=1.1$ is not considered in this example.

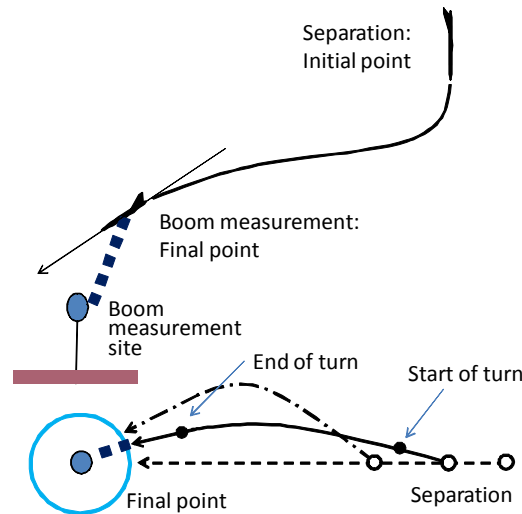


Figure 12. Image of range control by turn maneuver.

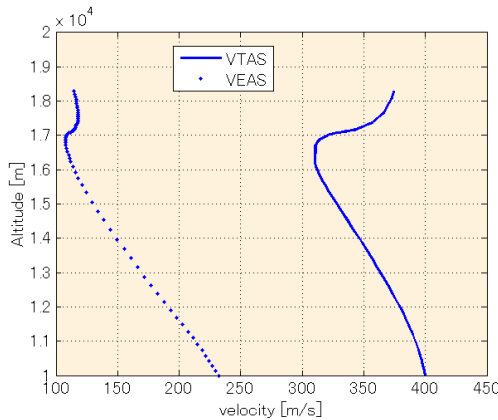


Figure 13a) Range control by turn maneuver, velocity.

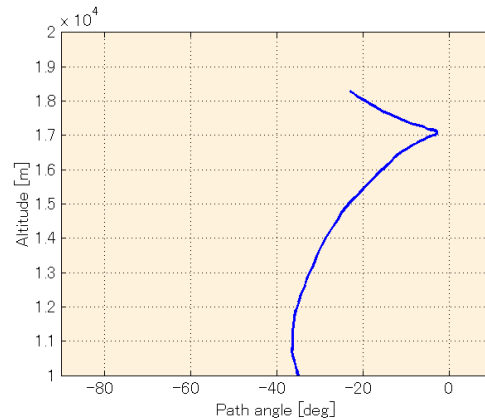


Figure 13b) Range control by turn maneuver, path angle.

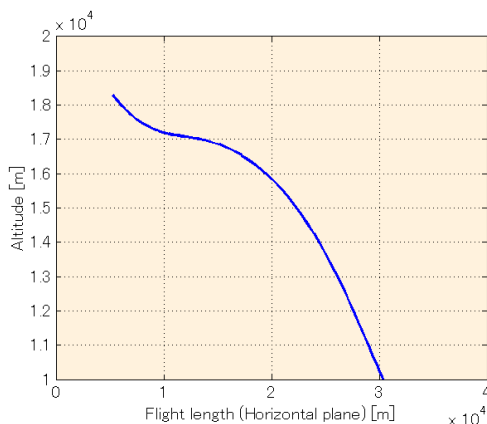


Figure 13c) Range control by turn maneuver, flight path.

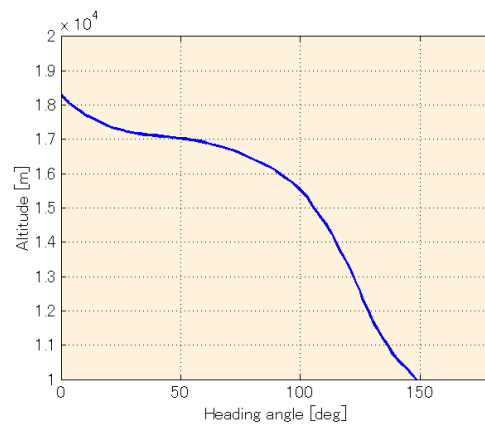


Figure 13d) Range control by turn maneuver, heading angle.

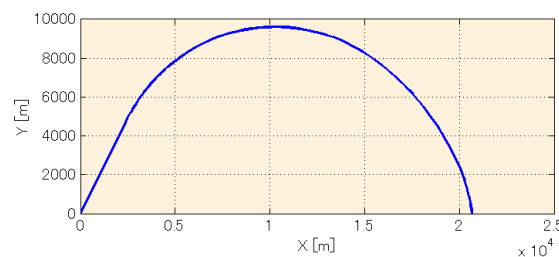


Figure 13e) Range control by turn maneuver, horizontal plane flight trajectory.

IX. Acceleration techniques in numerical computation

Computational time for each trajectory optimization by a modest performance personal computer is less than several hours, which is basically acceptable in the practical design process. Computational time, however, is always a trade-off against accuracy or grid size in DP. In other words, number of grid points are determined by tolerable computational time with given hardware resources. Therefore, reduction effort is always required. Using higher performance computers is definitely one candidate, but there are some other possibilities in the DP calculation. Two different techniques are proposed for the design tool. One is search space reduction method for each transition, and the other is iterative partial space DP calculation method. The former method is effective when the evaluation space in each transition can be explicitly limited by the properties of plant dynamics. Actually, in application examples of

this paper it is used as explained in each result. As the condition is often applicable in general, it is useful with a little effort of estimating the limit of possible transition.

The other method is a sort of combination of gradient method and DP calculation, therefore, it needs an initial trajectory. The method does not search all the grid space from the beginning, but it starts with a partial space around a flyable initial trajectory. Computational time for the partial space DP is so much decreased due to the number of grid points reduction. After it obtained the optimal trajectory for the limited partial space, then it moves the search space around the new optimal trajectory. It iterates the process till it reaches the same optimal trajectory. This technique is used in the calculation of range adjustment by the change of weighted penalty. In general, the technique is effective for the application in which the global optimality is not a problem.

X. Conclusion

A trajectory optimization tool has been developed for the design of guidance and control laws of the D-SEND#2 low sonic boom flight experiment vehicle. Dynamic programming (DP) is adopted for its favorable nature, such as multiple inequality constraints are easy to be implemented, and the global optimum solution can be obtained without iterative calculation to be converged. Results obtained by the tool gave useful information to design engineers of the D-SEND#2 project.

The research started with a motivation of developing an easy-to-use trajectory optimization tool which is welcomed by engineers involved in flight vehicle's guidance, navigation and control law design. The following characteristics are to be satisfied for the purpose.

- 1) Solution can be obtained and it is accurate.
- 2) Computational time is reasonable, and its upper bound can be estimated.
- 3) No special skill or experience is necessary for setting parameters.
- 4) No special skill or experience is necessary for imposing equality and inequality constraints.

The reason why we started the research is that most design engineers don't feel that they presently have an easy-to-use and reliable trajectory optimization tool. They feel that numerical solution of trajectory optimization is so complex that other methods based on geometrical solution with a simplified model are preferable. This paper challenges this classical problem by using dynamic programming. Curse of dimensionality and quantization error are problems of the method in its nature. It is difficult to solve the problem, but this paper can expand its application area to some extent, especially it proposes a technique to solve the relative order problem, which demands fine grid and large computational time. As the tendency of decreasing computational cost for over a half century does not seem to stop, DP will be a more widely used method for trajectory optimization in flight vehicle design in the future.

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